

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination

December 2013

CL105 Mechanics of Structures

Date: 07.12.2013, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Locate the centroid of a lamina shown in Figure 1. [07]

Q - 2 (a) Attempt any Two. [04]

- (i) Convert 2×10^5 to kN/m^2 .
- (ii) Two tensile forces 50kN and 40kN acting at a point with angle 60° between them. Find magnitude and direction of the resultant.

(iii) State principle of superposition.

(b) Attempt any Two. [10]

- (i) Find the magnitude of two forces such that if they act at right angles, their resultant is $\sqrt{40}$ kN, but if they act at 60° , their resultant is $\sqrt{52}$ kN.

Note: $\sqrt{}$: Square Root

- (ii) Find the angle between two equal forces of magnitude P when their resultant is 1.5P. What is the maximum possible value of resultant?

(iii) Determine the resultant of a co-planar con-current force system if following forces are acting at a point:

1. 150 kN tensile making an angle of 30° with +X axis.
2. 200 kN tensile making an angle of 120° with +X axis.
3. 80 kN tensile making an angle of 240° with +X axis.
4. 200 kN tensile making an angle of 310° with +X axis.

Q - 3 (a) Write four characteristics of a couple. [04]

(b) Attempt any Two. [10]

- (i) Find the moment of force 30kN about point 'O' shown in Figure 2.
- (ii) A square ABCD of side 1m is acted upon by the force system shown in Figure 3. Find resultant of the system with respect to point A.
- (iii) An electric lamp of 50 N weight is suspended by two wires of 4m and 3m length. The

horizontal distance between two fixed points is 5m from which two wires are suspended. Find tension in both wires.

SECTION – II

- Q - 4 A uniform ladder rests with one end on a horizontal floor and the other end against a vertical wall. Co-efficient of friction for the floor and the wall are 0.4 and 0.25 respectively. Find the inclination of the ladder when it is about to slip. [07]
- Q - 5 (a) Explain stress-strain curve of a ductile material when subjected to a tensile loading up to failure. [04]
- (b) **Attempt any Two.** [10]
- (i) A circular bar 3m long, uniformly tapers from 50mm diameter to 25mm diameter. Find the magnitude of the force which will deform the bar by 0.8mm. $E = 200 \text{ GPa}$.
- (ii) A load of 2000kN is applied on a short concrete column 500mm x 500mm, reinforced with 4, 10mm diameter steel bars. Find stresses in steel and concrete if $E_s = 15E_c$.
- (iii) A steel rod, 20mm diameter and 200mm long is heated through 100°C and at the same time subjected to an axial pull of 'P'. If the total extension of the rod is 0.30 mm, calculate the magnitude of pull 'P'. Consider $E = 200\text{GPa}$ and co-efficient of linear expansion is $12 \times 10^{-6}/^\circ\text{C}$.
- Q - 6 (a) A stepped bar loaded as shown in Figure 4. Calculate the stresses in each part and total change in length of bar. Consider $E_s = 200\text{GPa}$, $E_b = 80 \text{ GPa}$, $E_c = 100 \text{ GPa}$. [07]
- (b) **Attempt any One.** [07]
- (i) A man and a boy carry 30kg weight by means of a beam 2m long. Self weight of the beam is 10kg. Where the weight should be placed so that the man carries twice the weight carried by the boy?
- (ii) A beam ABC is 6m long. End A is hinged and end B is roller. Length of AB is 4m and BC is 2m. BC is an overhanging portion. Loads of 5kN and 8kN are acting at 1m and 3m from end A. Span BC is loaded with a UDL of 2 kN/m. Calculate the reactions at A and B.

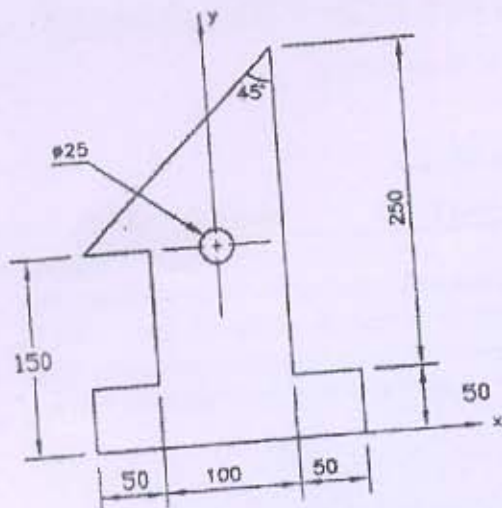


Figure 1

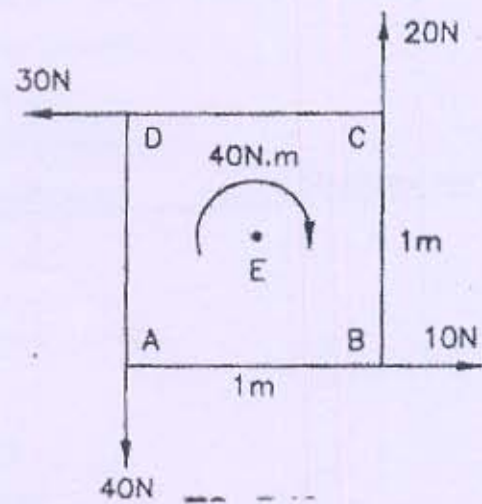


Figure 3

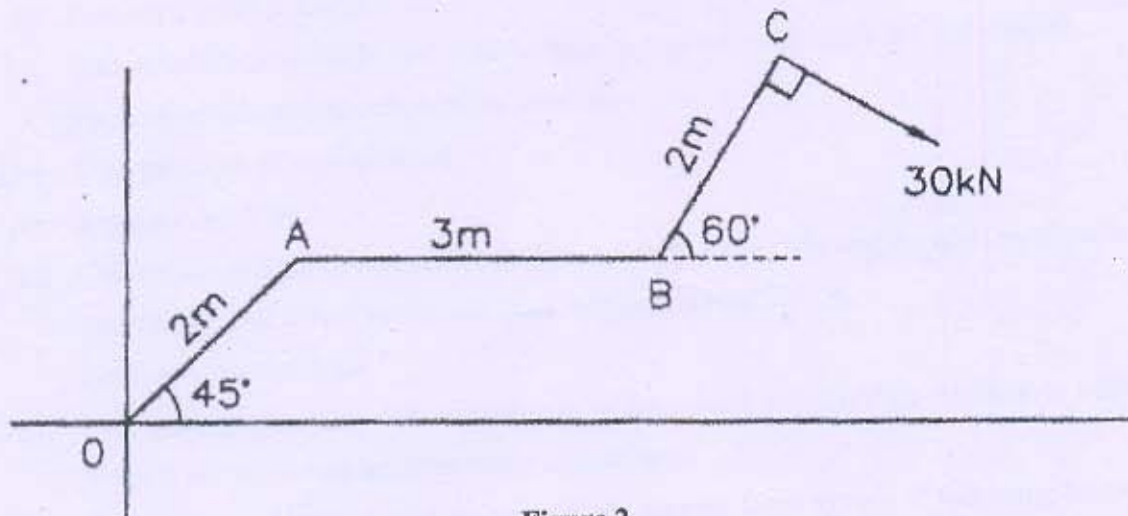


Figure 2

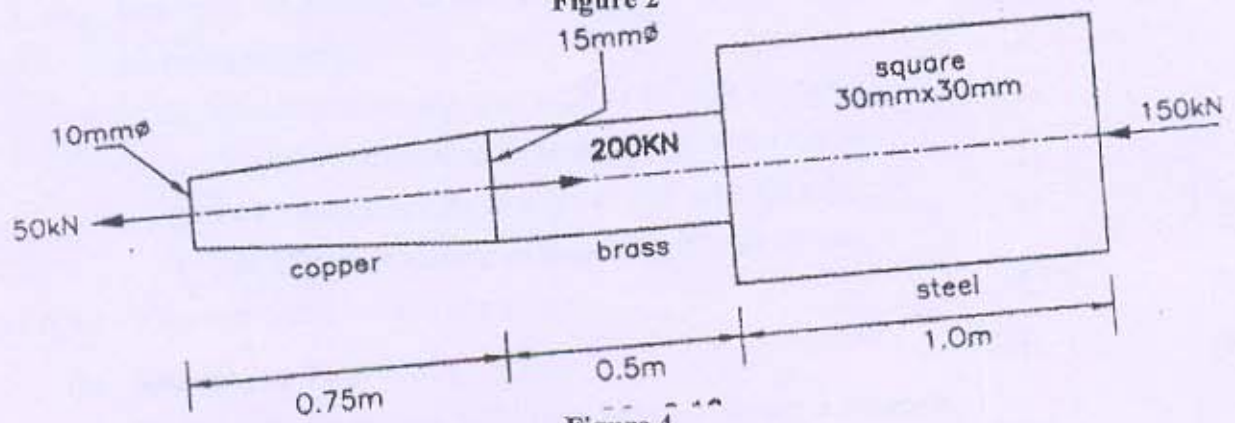


Figure 4